STUDENT REPORT

DETAILS

Name

THANGADIPALLY ANVITHA

Roll Number

20R01A6753

EXPERIMENT

Title

WEEK 2

Description

2. Demonstration of Association rule process on dataset contactlenses.arff using apriori algorithm.

Aim: This experiment illustrates some of the basic elements of asscociation rule mining using WEKA. The sample dataset used for this example is contactlenses.arff
Step1: Open the data file in Weka Explorer. It is presumed that the required data fields have been discretized. In this example it is age attribute.
Step2: Clicking on the associate tab will bring up the interface for association rule algorithm.
Step3: We will use apriori algorithm. This is the default algorithm.
Step4: Inorder to change the parameters for the run (example support, confidence etc) we click on the text box immediately to the right of the choose button.
Description:
The Apriori Algorithm: Finding Frequent Itemsets Using Candidate Generation
Apriori is a seminal algorithm proposed by R. Agrawal and R. Srikant in 1994 for mining frequent itemsets for Boolean association rules. The name of the algorithm is based on the fact that the algorithm uses *prior knowledge* of frequent itemset properties. Apriori employs an iterative approach known as a *level-wise* search, where *k*-itemsets are used to explore (*k*+1)-itemsets.
Apriori property: All nonempty subsets of a frequent itemset must also be frequent.
It has two steps.
1) Join step
2) Prune step
Procedure:
Step 1: Load appropriate dataset into weka
Step 2: Select associate tab and select apriori algorithm

Procedure and Overvation:

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and R.srikant in 1994 for mining frequent itemsets for boolean
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level-wise search ,where k-itemsets are used to explore (k+1)
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Apriori property: all nonempty subsets of a frequent itemset
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1)join step

2) Prune step

Procedure :

Step 1: load appropriate data set into weka

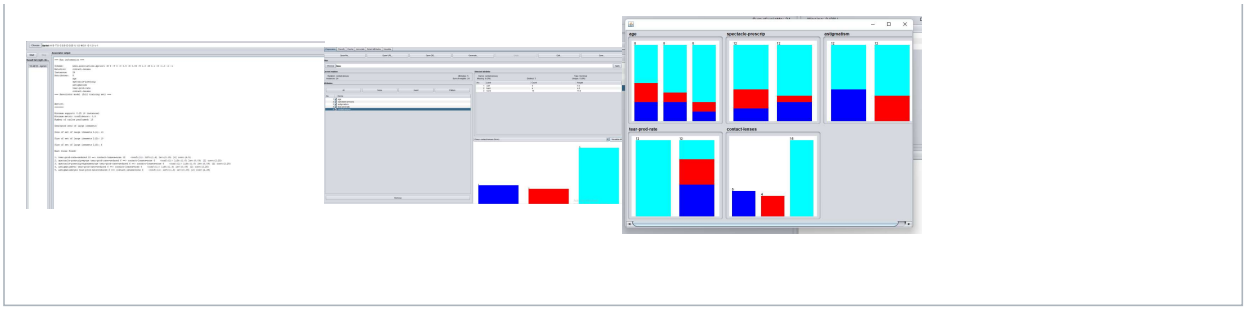
Step 2: select associate tab and select apriori algorithm

Apriori algorithm with 0.25 support and 0.9 confidence values

Dataset contactlenses.arff

```
@relation contact-lenses

@attribute age {young,pre-presbyopic,presbyopic}

@attribute spectacle-prescrip {myope,hypermetrope}

@attribute astigmatism {no,yes}

@attribute tear-prod-rate {reduced,normal}
```

```
@attribute contact-lenses {soft,hard,none}

@data

%

% 24 instances

%
young,myope,no,reduced,none

young,myope,no,normal,soft


young,myope,yes,reduced,none

young,myope,yes,normal,hard

young,hypermetrope,no,reduced,none


young,hypermetrope,no,normal,soft


young,hypermetrope,yes,reduced,none

young,hypermetrope,yes,normal,hard

pre-presbyopic,myope,no,reduced,none


pre-presbyopic,myope,no,normal,soft

pre-presbyopic,myope,yes,reduced,none


pre-presbyopic,myope,yes,normal,hard


pre-presbyopic,hypermetrope,no,reduced,none

pre-presbyopic,hypermetrope,no,normal,soft

pre-presbyopic,hypermetrope,yes,reduced,none


pre-presbyopic,hypermetrope,yes,normal,none


presbyopic,myope,no,reduced,none


presbyopic,myope,no,normal,none


presbyopic,myope,yes,reduced,none

presbyopic,myope,yes,normal,hard

presbyopic,hypermetrope,no,reduced,none


presbyopic,hypermetrope,no,normal,soft


presbyopic,hypermetrope,yes,reduced,none


presbyopic,hypermetrope,yes,normal,none


%

%
```



VIVA RESPONSES

Q) State whether True or False: Data warehouse is generally updated in real-time.

False

Q) What does OLTP stand for?

Online transaction processing

Q) Identify the correct option which defines Datamart.

A subgroup of data warehouse

Q) Where is data warehousing used?

Decision support system

Q) Identify the options below that a data warehouse can include.

All of the above